

CAMBRIDGE HARBOR, MD, USA

WMO: 997297

Lat: 38.5739N

Lon: 76.0686W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -5.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 12.9	(k) -2.6	(l) 11.8	(m) -2.5	(n) 4.0	(o) 310	(p) 0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.9	(c) 32.8	(d) N/A	(e) 31.6	(f) N/A	(g) 30.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.3	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.6	(b) 9.8	(c) 8.4	(e) -9.8	(f) 35.0	(g) 2.7	(h) 1.2	(i) -11.8	(j) 35.9	(k) -13.4	(l) 36.6	(m) -14.9	(n) 37.3	(o) -16.9	(p) 38.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	2.4	3.4	7.2	14.2	19.2	24.1	26.5	25.4	21.9	16.2	9.8	5.9		
	DBStd	9.31	5.05	4.92	4.70	4.33	3.83	3.02	2.31	2.21	3.02	4.04	4.09	4.52		
	HDD10.0	757	241	193	114	11	0	0	0	0	0	3	52	144		
	HDD18.3	2171	494	418	346	138	36	2	0	0	4	90	256	386		
	CDD10.0	2487	5	8	27	136	287	422	511	477	357	195	46	17		
	CDD18.3	859	0	0	1	13	64	174	253	219	112	24	0	0		
	CDH23.3	6644	0	1	4	87	349	1312	2443	1756	611	81	0	0		
	CDH26.7	1986	0	0	0	19	69	385	864	507	131	11	0	0		
(14) Wind	WSAvg	3.4	3.2	3.3	3.8	3.8	3.2	3.1	2.7	2.9	3.4	3.8	3.9	3.8		
(15) Precipitation	PrecAvg	1183	92	81	105	89	111	91	124	120	97	83	90	92		
	PrecMax	1698	192	198	204	180	249	297	314	273	264	203	217	226		
	PrecMin	805	11	14	46	16	21	14	35	13	14	1	12	15		
	PrecStd	221	38	45	43	38	61	61	64	78	63	49	53	56		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	19.4	23.7	29.1	30.8	33.8	35.3	33.9	31.7	28.1	21.5	20.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	15.1	16.2	19.5	25.9	28.7	31.7	33.2	31.9	29.9	25.4	19.6	17.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	12.6	13.6	16.8	23.2	26.9	30.3	31.8	30.7	28.3	23.8	17.9	15.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	9.8	10.5	14.1	21.0	25.3	28.9	30.7	29.5	26.7	22.2	16.1	12.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.9	7.0	7.5	8.4	8.0	7.1	6.9	6.6	6.7	6.8	6.7	6.2		
	5% DB	MCDBR	11.6	11.6	11.8	11.4	9.2	8.7	8.1	7.8	8.2	8.4	8.9	9.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.320	0.356	0.396	0.441	0.489	0.516	0.508	0.413	0.360	0.327	0.312		
	taud		2.485	2.459	2.406	2.312	2.219	2.132	2.069	2.077	2.319	2.469	2.521	2.528		
	Ebn at Noon		866	908	903	882	845	802	775	770	831	850	844	842		
	Edh at Noon		79	93	108	126	141	154	163	158	117	91	76	72		
(44) All-Sky Solar Radiation	RadAvg		2.17	3.00	4.00	5.08	5.76	6.33	6.17	5.51	4.47	3.37	2.44	1.90		
	RadStd		0.17	0.32	0.33	0.37	0.66	0.48	0.29	0.32	0.46	0.36	0.21	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+84

Nomenclature: See separate page